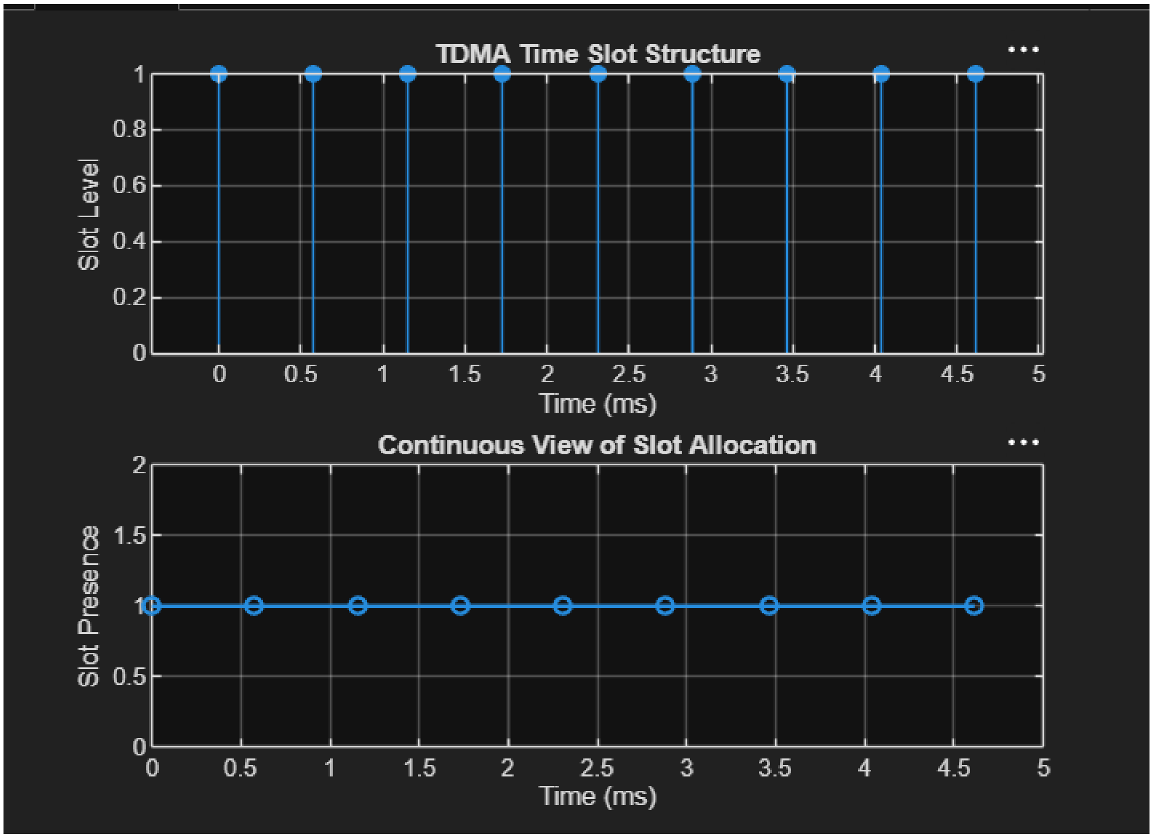
Ex.No:2  Implementing TDMA Slot Allocation in GSM. Date

Aim: To implement TDMA (Time Division Multiple Access) slot allocation in GSM using MATLAB and analyze system performance in terms of time slot duration, frame utilization, and number of allocated slots.

## Objective 1: Matlab Code and Output

```
clc; clear; close all;
FrameTime = 4.615; Slots = 8;
SlotDuration = FrameTime/Slots;

t = 0:SlotDuration:FrameTime;
figure
subplot(2,1,1)
stem(t,ones(size(t)),'filled')
title('TDMA Time Slot Structure')
xlabel('Time (ms)')
ylabel('Slot Level')
grid on
subplot(2,1,2)
plot(t,ones(size(t)),'-o','LineWidth',1.5)
title('Continuous View of Slot Allocation')
xlabel('Time (ms)')
ylabel('Slot Presence')
grid on
```

Output:




• The graph shows discrete time slots arranged in a GSM TDMA frame structure with equal spacing.

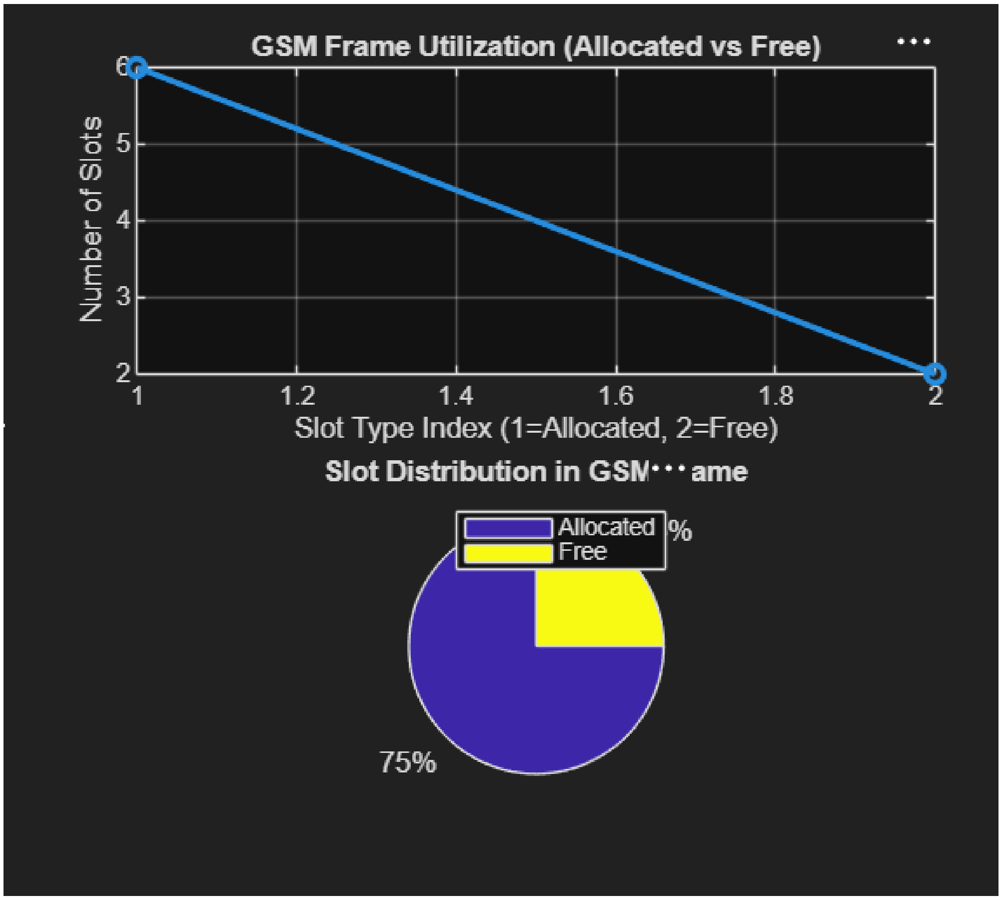
• It confirms fixed time division where each slot is allocated a specific time interval without overlap.

## Objective 2: Matlab Code and Output

```
clc; clear; close all;
TotalSlots = 8;
AllocatedSlots = 6;
FreeSlots = TotalSlots - AllocatedSlots;

Utilization = (AllocatedSlots/TotalSlots)*100;
disp('Frame Utilization (%)')
```

```
disp(Utilization)
figure
% Graph 1: Simple Line Plot (SAFE - no bar function)
subplot(2,1,1)
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
title('GSM Frame Utilization (Allocated vs Free)')
xlabel('Slot Type Index (1=Allocated, 2=Free)')
ylabel('Number of Slots')
grid on
% Graph 2: Pie Chart (SAFE)
subplot(2,1,2)
pie([AllocatedSlots FreeSlots])
title('Slot Distribution in GSM Frame')
legend('Allocated','Free')
```

Output :



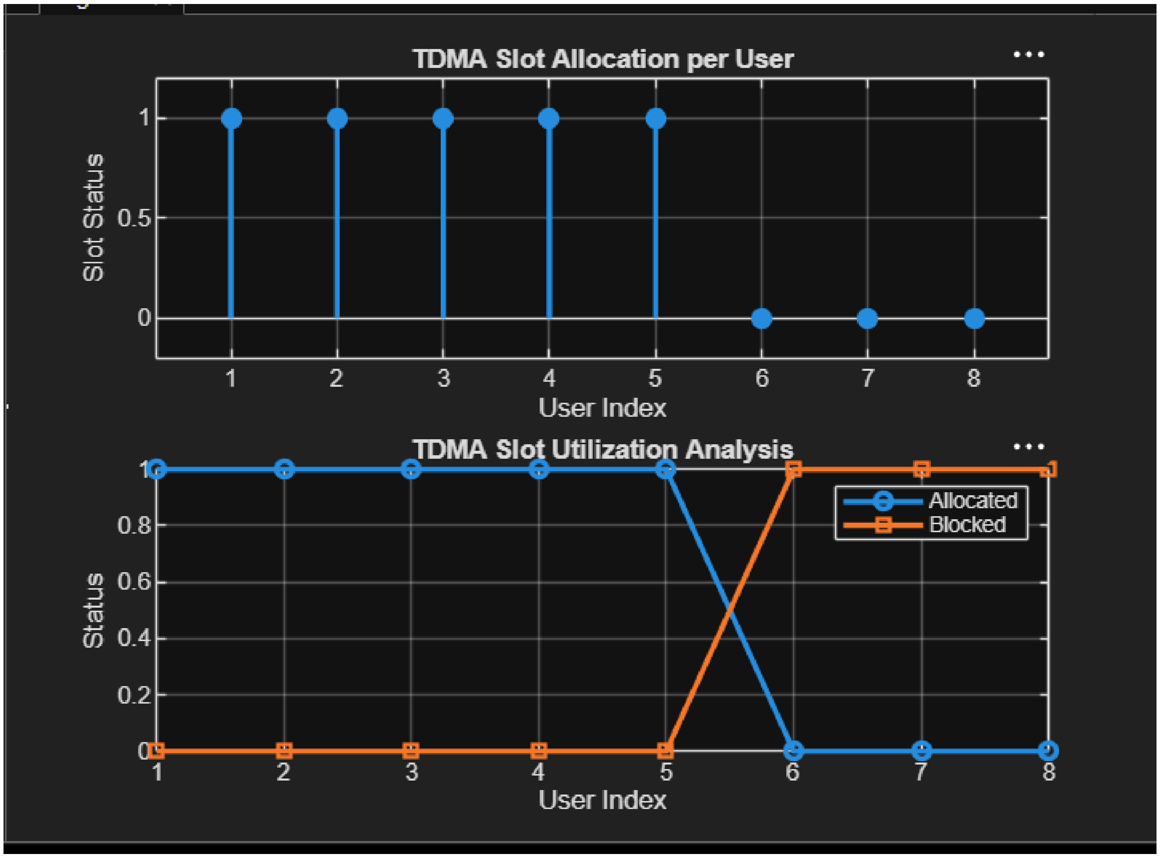

- The plot shows that most of the TDMA frame is allocated to active users, indicating highb resource usage.
- The pie chart clearly represents that 75% of slots are utilized while 25% remain

## Objective 3: Matlab Code and Output

```
clc; clear; close all;


Users = 1:8; % Total users

TotalSlots = 5; % Available slots

Alloc = zeros(1,8); % Initialize allocation

Alloc(1:TotalSlots) = 1; % Assign slots to first users

Blocked = 1 - Alloc; % Remaining users blocked

disp('User Wise Slot Allocation')

disp(table(Users',Alloc','VariableNames',{'User','Status(1=Allocated,0=Blocked)'})
```

```
) figure
% Graph 1: Allocation per user
subplot(2,1,1)
stem(Users,Alloc,'filled','LineWidth',2)
title('TDMA Slot Allocation per User')
xlabel('User Index')
ylabel('Slot Status (1=Allocated, 0=Blocked)')
ylim([-0.2 1.2])
grid on
% Graph 2: System view (safe alternative to bar)
subplot(2,1,2)
plot(Users,Alloc,'-o','LineWidth',2)
hold on
plot(Users,Blocked,'-s','LineWidth',2)
title('TDMA Slot Utilization Analysis')
xlabel('User Index')
ylabel('Status')
legend('Allocated','Blocked')
grid on
```

OUTPUT




- The first graph shows which users are allocated time slots and which are blocked due to limitet resources.
- The second graph compares allocated and blocked users, clearly showing TDMA resource limitation.